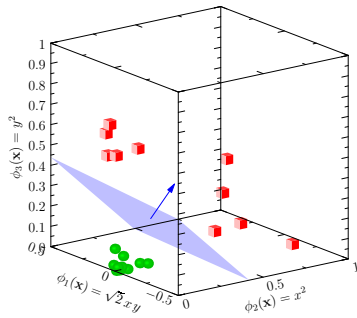




The Kernel Trick, SVMs, Regression

## SVM Kernels

- SVM Kernels are functions of two variables that can be factorised

$$K(\mathbf{x}, \mathbf{y}) = \langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle = \sum_i \phi_i(\mathbf{x}) \phi_i(\mathbf{y})$$

- where  $\phi(\mathbf{x}) = (\phi_1(\mathbf{x}), \phi_2(\mathbf{x}), \dots)^\top$  and  $\phi_i(\mathbf{x})$  are real valued functions of  $\mathbf{x}$
- $K(\mathbf{x}, \mathbf{y})$  will be positive semi-definite (because it is an inner-product)
- Furthermore, any positive semi-definite function will factorise
- This factorisation is not always obvious (we return to this later)

## Classifying New Data

- Having trained the SVM we now have to use it
- Given a new input  $\mathbf{x}$  we decide on the class

$$y = \text{sgn}(\langle \mathbf{w}, \phi(\mathbf{x}) \rangle - b) \quad \text{but} \quad \mathbf{w} = \sum_{k=1}^m \alpha_k y_k \phi(\mathbf{x}_k)$$

- In the dual representation this becomes

$$\text{sgn} \left( \sum_{k=1}^m \alpha_k y_k K(\mathbf{x}_k, \mathbf{x}) - b \right)$$

where we only need to sum over the non-zero  $\alpha_k$  (i.e. the support vectors SVs)

## Recap on Eigen Systems

- Recall for a symmetric ( $n \times n$ ) matrix  $\mathbf{M}$  an eigenvector,  $\mathbf{v}$

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$$

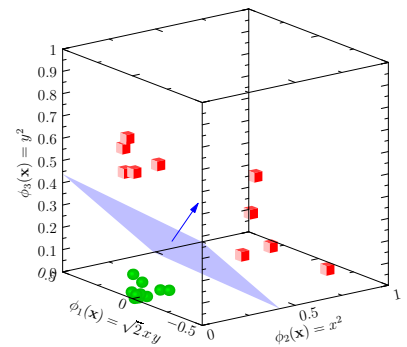
- There are  $n$  independent eigenvectors  $\mathbf{v}^{(i)}$  with real eigenvalues  $\lambda^{(i)}$
- The eigenvectors are orthogonal so that  $\mathbf{v}^{(i)\top} \mathbf{v}^{(j)} = 0$  if  $i \neq j$
- Forming a matrix of eigenvectors  $\mathbf{V} = (\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \dots, \mathbf{v}^{(n)})$  the matrix satisfies

$$\mathbf{V}^\top \mathbf{V} = \mathbf{V} \mathbf{V}^\top = \mathbf{I}$$

- Such matrices are said to be orthogonal

## Outline

- The Kernel Trick
- Positive Semi-Definite Kernels
- Kernel Properties
- Beyond Classification



## Dual Form

- Recall that the dual problem for an SVM is

$$\max_{\alpha} \sum_{k=1}^m \alpha_k - \frac{1}{2} \sum_{k,l=1}^m \alpha_k \alpha_l y_k y_l \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$$

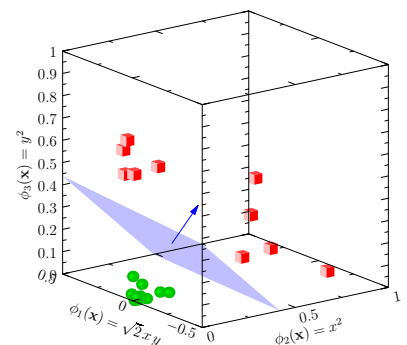
- subject to  $\sum_{k=1}^m y_k \alpha_k = 0$  and  $0 \leq \alpha_k \leq C$
- But since  $K(\mathbf{x}_k, \mathbf{x}_l) = \langle \phi(\mathbf{x}_k), \phi(\mathbf{x}_l) \rangle$  the dual problem becomes

$$\max_{\alpha} \sum_{k=1}^m \alpha_k - \frac{1}{2} \sum_{k,l=1}^m \alpha_k \alpha_l y_k y_l K(\mathbf{x}_k, \mathbf{x}_l)$$

- This is the **kernel trick**—we never have to compute  $\phi(\mathbf{x})$

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## Eigen Decomposition

- From the eigenvalue equation  $\mathbf{M}\mathbf{v}^{(k)} = \lambda^{(k)} \mathbf{v}^{(k)}$

$$\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{\Lambda} \quad \text{where} \quad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

- Multiplying on the right by  $\mathbf{V}^\top$  we get

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top = \sum_{k=1}^n \lambda^{(k)} \mathbf{v}^{(k)} \mathbf{v}^{(k)\top}$$

Or

$$M_{ij} = \sum_{k=1}^n \lambda^{(k)} v_i^{(k)} v_j^{(k)} = \sum_{k=1}^n u_i^{(k)} u_j^{(k)} = \langle \mathbf{u}_i, \mathbf{u}_j \rangle$$

$$u_i^{(k)} = \sqrt{\lambda^{(k)}} v_i^{(k)}$$

## Eigenfunctions

- By analogy for a symmetric function of two variables we can define an *eigenfunction*

$$\int K(x, y) \psi(y) dy = \lambda \psi(x)$$

- In general there will be a denumerable set of eigenfunctions  $\psi^{(k)}(x)$  where

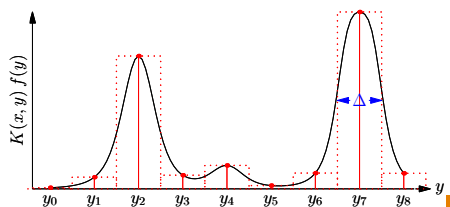
$$K(x, y) = \sum_k \lambda^{(k)} \psi^{(k)}(x) \psi^{(k)}(y)$$

- This is known as Mercer's theorem

## Linear Operators

- Recall a linear function  $\mathcal{T}[f(x)]$  can be represented by a kernel

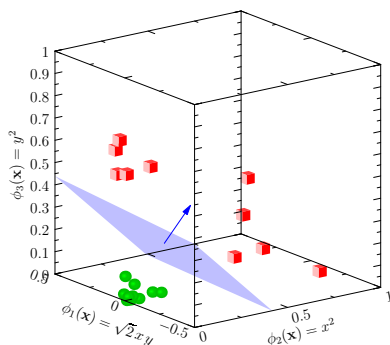
$$\mathcal{T}[f(x)] = \int_{y \in \mathcal{I}} K(x, y) f(y) dy \approx \Delta \sum_{j=1}^n K(x, y_j) f(y_j)$$



This is just a matrix equation with  $M_{ij} = \Delta K(x_i, y_j)$

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## Properties of Positive Semi-Definiteness

- Since

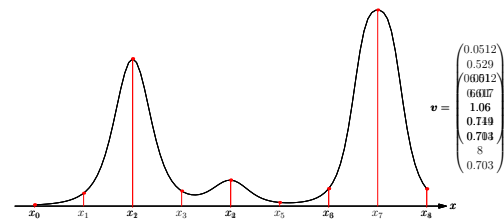
$$K(x, y) = \langle \phi(x), \phi(y) \rangle$$

- An immediate consequence is that for any function  $f(x)$

$$\begin{aligned} \int f(x) K(x, y) f(y) dx dy &= \int f(x) \langle \phi(x), \phi(y) \rangle f(y) dx dy \\ &= \left\langle \int f(x) \phi(x) dx, \int f(y) \phi(y) dy \right\rangle \\ &= \left\| \int f(x) \phi(x) dx \right\|^2 \geq 0 \end{aligned}$$

## Limit Process

- Consider sampling a function at a set of points



- In the limit where the number of sample points goes to infinity the vector more closely approximates a function
- The  $k^{th}$  component of the vector corresponds to  $f(x_k)$

## SVM Kernels

- If we define  $\phi^{(k)}(x) = \sqrt{\lambda^{(k)}} \psi^{(k)}(x)$  then

$$K(x, y) = \sum_k \lambda^{(k)} \psi^{(k)}(x) \psi^{(k)}(y) = \sum_k \phi^{(k)}(x) \phi^{(k)}(y)$$

- This is the definition of a SVM kernel we started with
- Note that for  $\phi^{(k)}(x)$  to be real  $\lambda^{(k)} \geq 0$  for all  $k$
- If  $\lambda^{(k)} < 0$  then  $\langle \phi(x), \phi(x) \rangle = \|\phi(x)\|^2$  might be negative and "distance" between points in the extended feature space can be negative
- If we use a kernel that isn't positive semi-definite then the Hessian of the dual objective function will not be negative semi-definite and there will be a maximum where  $\alpha$  diverges

## Positive Semi-Definite Kernels

- Kernels (or matrices) that have eigenvalues  $\lambda^{(k)} \geq 0$  are called positive semi-definite
- (If the eigenvalues are strictly positive  $\lambda^{(k)} > 0$  the kernels or matrices are called positive definite)
- Positive semi-definite kernels can always be decomposed into a sum of real functions

$$K(x, y) = \langle \phi(x), \phi(y) \rangle$$

## Positive Semi-Definiteness

- The following statements are equivalent

- $K(x, y)$  is positive semi-definite (written  $K(x, y) \succeq 0$ )
- The eigenvalues of  $K(x, y)$  are non-negative
- The kernel can be written

$$K(x, y) = \langle \phi(x), \phi(y) \rangle$$

where the  $\phi^{(k)}(x)$ 's are real functions

- For any real function  $f(x)$

$$\int f(x) K(x, y) f(y) dx dy \geq 0$$

## Adding Kernels

- We can construct SVM kernels from other kernels

- If  $K_1(x, y)$  and  $K_2(x, y)$  are valid kernels then so is  $K_3(x, y) = K_1(x, y) + K_2(x, y)$

$$\begin{aligned} Q &= \int f(x) K_3(x, y) f(y) dx dy \\ &= \int f(x) (K_1(x, y) + K_2(x, y)) f(y) dx dy \\ &= \int f(x) K_1(x, y) f(y) dx dy + \int f(x) K_2(x, y) f(y) dx dy \geq 0 \end{aligned}$$

- If  $K(x, y)$  is a valid kernel so is  $cK(x, y)$  for  $c > 0$

## Exponentiating Kernels

- If  $K(x, y)$  is a valid kernel so is  $K^n(x, y)$  (by induction)
  - ★ Assume  $K(x, y) \succeq 0$  this satisfies base case
  - ★ If  $K^{n-1}(x, y) \succeq 0$  then

$$K^n(x, y) = K^{n-1}(x, y) K(x, y) \succeq 0$$

- and  $\exp(K(x, y))$  is also a valid kernel since

$$e^{K(x, y)} = \sum_{i=1}^{\infty} \frac{1}{i!} K^i(x, y) = 1 + K(x, y) + \frac{1}{2} K^2(x, y) + \dots$$

but each term in the sum is a kernel

## Other Kernels

- The success of SVMs has meant that researchers try to increase the area of application
- The condition that a SVM kernel must be positive semi-definite is quite restrictive
- There has been a cottage industry of researchers finding smart kernels for solving complicated problems
- The key to finding new kernels is to use the properties of kernels to build more complicated kernels from simpler ones

## Spectrum Kernel

- A simple way to compare documents is to collect a histogram of all occurrences of substrings of length  $p$
- This is known as a  $p$ -spectrum
- A  $p$ -spectrum kernel counts the number of common substrings

$$\begin{aligned} s &= \text{statistics} & \mathcal{S}_3(s) &= \{\text{sta}, \text{tat}, \text{ati}, \text{tis}, \text{ist}, \text{sti}, \text{tic}, \text{ics}\} \\ t &= \text{computation} & \mathcal{S}_3(t) &= \{\text{com}, \text{omp}, \text{mpu}, \text{put}, \text{uta}, \text{tat}, \text{ati}, \text{tio}, \text{ion}\} \end{aligned}$$

- $K(s, t) = 2$  ("tat" and "ati")

## Product of Kernels

- If  $K_1(x, y)$  and  $K_2(x, y)$  are valid kernels then so is  $K_3(x, y) = K_1(x, y) K_2(x, y)$

- Writing

$$K_1(x, y) = \sum_i \phi_i^{(1)}(x) \phi_i^{(1)}(y), \quad K_2(x, y) = \sum_j \phi_j^{(2)}(x) \phi_j^{(2)}(y)$$

then

$$\begin{aligned} K_3(x, y) &= \sum_{i,j} \phi_i^{(1)}(x) \phi_i^{(1)}(y) \phi_j^{(2)}(x) \phi_j^{(2)}(y) \\ &= \sum_{i,j} \phi_{ij}^{(3)}(x) \phi_{ij}^{(3)}(y) = \left\langle \phi^{(3)}(x), \phi^{(3)}(y) \right\rangle \end{aligned}$$

$$\text{where } \phi_{ij}^{(3)}(x) = \phi_i^{(1)}(x) \phi_j^{(2)}(x)$$

## RBF Kernel

- Now  $x^T y = \langle x, y \rangle$  is a valid kernel because it is an inner product of functions  $\phi(x) = x$
- For  $\gamma > 0$  we have  $2\gamma x^T y \succeq 0$
- Thus  $\exp(2\gamma x^T y) \succeq 0$
- If  $K(x, y) \succeq 0$  then  $g(x) K(x, y) g(y) \succeq 0$

$$\int f(x) g(x) K(x, y) g(y) f(y) dx dy = \int h(x) K(x, y) h(y) dx dy \succeq 0$$

where  $f(x) g(x) = h(x)$

$$e^{-\gamma x^T x} e^{2\gamma x^T y} e^{-\gamma y^T y} = e^{-\gamma \|x-y\|^2} \succeq 0$$

## String Kernels

- One area where SVMs were very important is in document classification
- This requires comparing strings
- There are a large number of kernels developed to do this

## All Subsequences Kernel

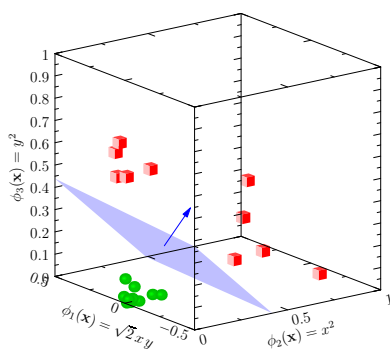
- A more sophisticated kernel is to count all of the common subsequences that occur in two documents
- Naively this would take an exponential amount of time to compute
- Using clever dynamic-programming techniques this can be done relatively efficiently
- This can even be extended to include sub-sequence matches with possible gaps between words

- String kernels for comparing subsequences are used in bioinformatics
- Kernels have been developed for comparing trees (e.g. for computer program evaluation, XML, etc.)
- Kernels have also been developed for comparing graphs (e.g. for comparing chemicals based on their molecular graph)

- In an attempt to build kernels that capture more domain knowledge, kernels are constructed from other learning machines
- An example of this are “Fisher kernels” whose features come from an Hidden Markov Model (HMM) trained on the data
- These tend to have better discriminative power than the underlying model (HMM), and has a better feature set than a SVM using a generic kernel

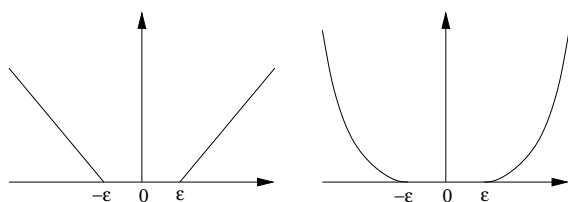
## Outline

1. The Kernel Trick
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## Error Functions

- Can introduce slack variables with different errors



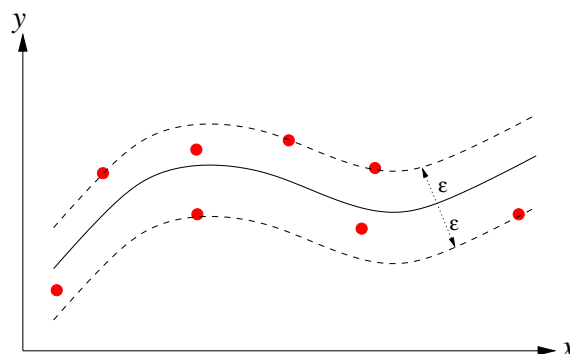
- This can be transformed to a quadratic programming problem

## Kernel Methods

- Kernel methods where we project into an extended feature space are used with other linear algorithms
  - ★ Kernel Fisher discriminant analysis (KFDA)
  - ★ Kernel principle component analysis (KPCA)
  - ★ Kernel canonical correlation analysis (KCCA)
  - ★ Gaussian Processes
- These are also extremely powerful machine learning algorithms

## Regression with Margins

- SVMs can be modified to perform regression



## Ridge Regression Using Kernels

- We can also solve regression problems without using margins
- To solve a regression problem once again the problem is set up as a quadratic programming problem

$$\min_w \lambda \|w\|^2 + \sum_{i=1}^m (y_i - w^T \phi(x_i))^2$$

- the  $\|w\|^2$  is a regularisation term
- By assuming  $w = \sum_i \alpha_i \phi(x_i)$  we obtain a quadratic equation for the  $\alpha_i$ 's which we can solve

## Summary

- SVMs require a positive definite kernel function
- These can be built from simpler function
- There was a cottage industry of people creating new kernels for different application
- SVMs are just one example of a host of machine that
  - ★ use the kernel trick
  - ★ often use linear constraints
  - ★ tend to be convex optimisation problems